

Programme Name: B.Tech *CSE AU*
 Course Name : Data Structures and Algorithms
 Course Code : CSEG1043
 Nos. of page(s) : 2
 Calculator allowed: No

Instructions: Please attempt according to the time provided and given weightage.

SECTION A

(30 Marks) 5 Questions

Attempt all questions (Not More than 50 words)

S. No.		Marks	CO
Q 1	"The name of an array stores the base address of the array." Explain this statement with the help of a small example.	6	1
Q 2	Explain the concept of an Anonymous Union with the help of a suitable code snippet.	6	1
Q 3	Consider the following incomplete program: <pre>int main(){ int a[] = {1, 2, 3, 4}; }</pre> void fun() { Complete the program so that the array is passed to the <u>function fun()</u> and its elements are <u>displayed inside the function body</u> .	6	1
Q 4	Write a C program to search for an element in an array and display its position if found. If the element is not present, display an appropriate message.	6	1
Q 5	Write a C program in which: 1. A function accepts two integer numbers as parameters. 2. The function calculates their sum and stores it in an integer variable named result. 3. The function returns the address of result to the main() function. 4. The main() function displays the sum using the returned address.	6	1

SECTION B

(45 Marks) 3 questions – (Q8 is having internal choice)

Q 6	Explain the concept of time complexity. Determine the time complexity of the following code: <pre>for(int i = 1; i <= n; i = i * 2){ printf("ABC"); }</pre> Also justify your answer with proper steps.	15	1
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Q 7	(a) Consider the following incomplete program: <pre> struct abc{ int a; char b; }; struct abc* fun(); int main(){ } struct abc* fun(){ }</pre> Complete the program to perform the following tasks: 1. Dynamically allocate memory for a structure variable inside the function fun(). 2. Assign suitable values to the members a and b inside the function fun(). 3. Return the address of the dynamically allocated structure to the main() function. 4. Display the values of the structure members in the main() function. (b) Explain why dynamic memory allocation is required while returning a structure pointer from a function.	15	1
Q 8	Write the program to perform the following operations in a Header Linked List: 1. Insert a node at the end 2. Delete a node from the end 3. Display the list OR Write a program to perform the following tasks: 1. Initialize an array in the main() function. 2. Pass the array to a function. 3. Inside the function, convert the array into a Singly Linked List by creating nodes dynamically. 4. Display the contents of the linked list.	15	2

SECTION C

(25 Marks) 1 Questions (Q 9 is having internal choice)

Q 9	Write the functions to perform the following operations in a Doubly Linked List: 1. Insert a node at the end. 2. Delete a node from the end. 3. Display the list in forward direction. 4. Display the list in backward direction. OR Write the functions to perform the following operations in a Circular Linked List: 1. Insert a node at the beginning 2. Insert a node at the end 3. Delete a node from the beginning 4. Traverse and display the list	25	2
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